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✓ No state motion changes

✓ No frequent collisions

✓ More predictable trajectories

⚡ SECTION 5: SPEED ANALYSIS

5.1 Two Critical Parameters

1. Inference Speed (Inferencing Time)

Time to process one frame = Detection + Tracking

Measured in: milliseconds (ms) or FPS

Lower ms = Better (faster processing)

What it includes:

- (a) Run object detector on frame
- (b) Execute tracker (associate + update)

Factors affecting:

- Detector complexity (YOLO vs Faster-RCNN)
- Tracker algorithm efficiency
- Hardware platform

2. Update Frequency Per Image

How frequently tracker updates internal state with new observations

Measured in: milliseconds (ms)

Lower ms = More frequent updates

✓ What it does: